

DATA CLEANING & ANALYTICS

PROJECT REPORT

DecodeLab Internship

E-Commerce Order Dataset Analysis

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Domain: Data Analytics
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Tool: Python (pandas, matplotlib) | Dataset: 1,200 Orders | 14 Features

1. Introduction

This report presents a complete data cleaning and analytics workflow performed on an E-Commerce order dataset containing 1,200 records. The primary objective was to identify, clean, and validate data quality issues using Python programming, and then derive meaningful insights from the cleaned data.

1.1 Dataset Overview

Attribute	Details
Dataset Name	E-Commerce Orders Dataset
Source Format	Microsoft Excel (.xlsx)
Total Records	1,200 orders

Attribute	Details
Total Features	14 columns
Date Range	January 2023 - June 2025
Key Domain	Online Retail / E-Commerce

1.2 Dataset Columns

Column Name	Data Type	Description
OrderID	String	Unique identifier for each order
Date	DateTime	Date when order was placed
CustomerID	String	Unique customer identifier
Product	String	Product name (7 categories)
Quantity	Integer	Number of items ordered
UnitPrice	Float	Price per unit in USD
ShippingAddress	String	Customer shipping address
PaymentMethod	String	Method used for payment
OrderStatus	String	Current status of the order
TrackingNumber	String	Shipment tracking code
ItemsInCart	Integer	Total items in cart at time of order
CouponCode	String	Discount coupon used (if any)
ReferralSource	String	How the customer found the platform
TotalPrice	Float	Final order total in USD

2. Methodology & Tools

The data cleaning process followed a structured pipeline with 8 distinct steps. Python was used as the primary programming language with the pandas library for all data manipulation tasks.

Libraries Used

Library	Version	Purpose
pandas	2.x	Data loading, cleaning, manipulation
matplotlib	3.x	Data visualization and charting
seaborn	0.12+	Statistical visualizations

Library	Version	Purpose
numpy	1.x	Numerical operations
openpyxl	3.x	Reading/writing Excel files

Data Cleaning Pipeline

- Step 1: Load dataset and perform initial inspection
- Step 2: Identify and handle missing values
- Step 3: Remove duplicate records
- Step 4: Correct data formats (dates, numeric precision, text casing)
- Step 5: Validate data ranges (negatives, zeros, future dates)
- Step 6: Audit data integrity (TotalPrice = Quantity x UnitPrice)
- Step 7: Standardize categorical values
- Step 8: Final verification and documentation

3. Data Cleaning - Step-by-Step

Step 1: Loading the Dataset

The dataset was loaded from an Excel file using pandas read_excel() function. The first five rows were displayed to understand the structure and content.

Python Code:

```
import pandas as pd

file_path = '/content/Dataset for Data Analytics (1).xlsx'
df = pd.read_excel(file_path)

print("Dataset loaded successfully. Displaying first 5 rows:")
display(df.head())
```

Output Preview (First 5 Rows):

OrderID	Date	Product	Qty	UnitPrice	Status	TotalPrice
ORD200000	2023-01-04	Monitor	5	570.62	Shipped	2853.10
ORD200001	2024-08-23	Phone	2	151.35	Shipped	302.70
ORD200002	2024-02-27	Tablet	5	550.68	Cancelled	2753.40
ORD200003	2023-10-15	Chair	1	273.19	Returned	273.19
ORD200004	2025-05-08	Printer	4	626.01	Delivered	2504.04

Step 2: Identify & Handle Missing Values

Missing value analysis revealed that the CouponCode column had 309 missing entries (25.75% of dataset). Since this is a categorical column, the mode (most frequent value) was used for imputation.

Python Code:

```
missing_values = df.isnull().sum()
missing_values = missing_values[missing_values > 0]

# Impute with mode
mode_coupon_code = df['CouponCode'].mode()[0]
df['CouponCode'] = df['CouponCode'].fillna(mode_coupon_code)
print(f'Imputed with mode: {mode_coupon_code}')
```

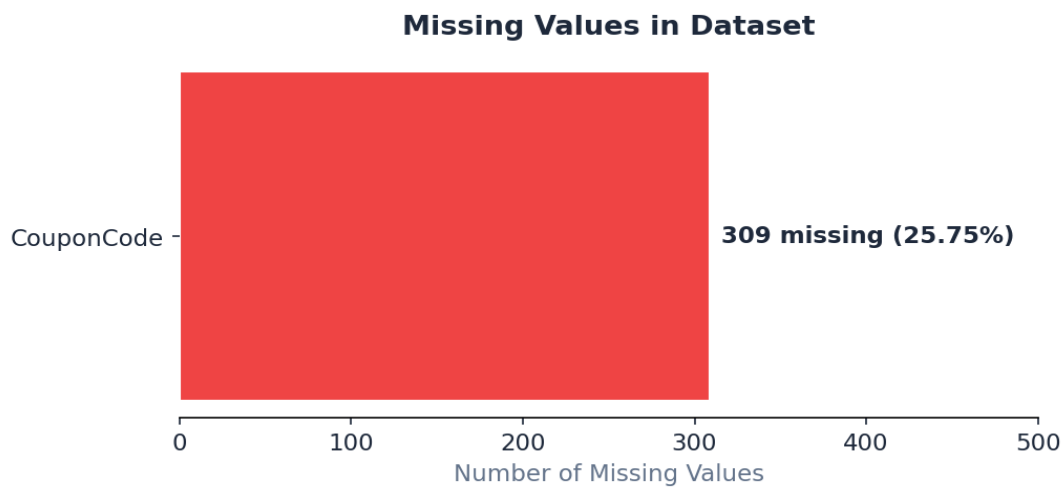


Figure 1: Missing values in CouponCode column (309 records imputed with mode = "FREESHIP")

Result: 309 missing CouponCode values were imputed with "FREESHIP" (the mode). After imputation, 0 missing values remain.

Step 3: Remove Duplicate Records

Duplicate OrderIDs were checked to ensure every order appears exactly once. Duplicates would lead to inflated metrics in analysis.

Python Code:

```
initial_rows = df.shape[0]
duplicate_order_ids = df[df.duplicated(subset='OrderID', keep=False)]

if not duplicate_order_ids.empty:
    df.drop_duplicates(subset='OrderID', keep='first', inplace=True)
```

```
else:  
    print('No duplicate OrderIDs found.')
```

Result: No duplicate OrderIDs were found. All 1,200 records are unique (0% error rate).

Step 4: Correct Data Formats

Three formatting corrections were applied: (1) Date column standardized to ISO 8601 format (YYYY-MM-DD), (2) Numeric columns rounded to 2 decimal places, (3) Text fields trimmed and title-cased.

Python Code:

```
# 1. Standardize date format  
df['Date'] = pd.to_datetime(df['Date'], errors='coerce').dt.strftime('%Y-%m-%d')  
  
# 2. Round numeric columns to 2 decimal places  
numeric_cols = ['Quantity', 'UnitPrice', 'TotalPrice']  
for col in numeric_cols:  
    df[col] = pd.to_numeric(df[col], errors='coerce').round(2)  
  
# 3. Trim and title-case text fields  
text_cols = ['Product', 'ShippingAddress', 'PaymentMethod',  
             'ReferralSource', 'OrderStatus']  
for col in text_cols:  
    df[col] = df[col].astype(str).str.strip().str.title()
```

Result: Dates standardized to YYYY-MM-DD. Numeric precision set to 2 decimal places. All text columns cleaned and properly cased.

Step 5: Validate Data Ranges

Numerical columns were validated for negative values and zero prices. The Date column was checked for future dates that could indicate data entry errors.

Python Code:

```
numerical_cols = ['Quantity', 'UnitPrice', 'TotalPrice', 'ItemsInCart']  
for col in numerical_cols:  
    negative_vals = df[df[col] < 0]  
    if not negative_vals.empty:  
        df.loc[df[col] < 0, col] = None # mark as NaN  
  
# Check future dates  
current_date = pd.to_datetime('today').normalize()  
future_dates = df[df['Date'] > current_date]  
# Flag and set to NaT if found
```

Result: No negative values found. No zero prices found. No invalid future dates found. All data ranges are valid.

Step 6: Audit Data Integrity

A critical integrity check was performed to verify that TotalPrice equals Quantity multiplied by UnitPrice for every row. Discrepancies indicate calculation errors or data corruption.

Python Code:

```
df['ExpectedTotalPrice'] = df['Quantity'] * df['UnitPrice']
tolerance = 0.01 # 2 decimal places accuracy

discrepancy_rows = df[
    abs(df['TotalPrice'] - df['ExpectedTotalPrice']) > tolerance
]

if not discrepancy_rows.empty:
    # Correct TotalPrice where mismatch found
    df.loc[..., 'TotalPrice'] = df['ExpectedTotalPrice']
df.drop(columns=['ExpectedTotalPrice'], inplace=True)
```

Result: No discrepancies found between TotalPrice and Quantity x UnitPrice. Financial data is 100% consistent.

Step 7: Standardize Categorical Values

Categorical columns (PaymentMethod, OrderStatus, ReferralSource) were standardized by mapping all variations to consistent labels using title-casing and replacement dictionaries.

Python Code:

```
# Standardize PaymentMethod
df['PaymentMethod'] = df['PaymentMethod'].str.title()

# Standardize OrderStatus
order_status_mapping = {
    'Delivered': 'Delivered', 'Shipped': 'Shipped',
    'Cancelled': 'Cancelled', 'Returned': 'Returned', 'Pending': 'Pending'
}
df['OrderStatus'] = df['OrderStatus'].replace(order_status_mapping)

# Standardize ReferralSource
df['ReferralSource'] = df['ReferralSource'].str.title()
```

Unique Values After Standardization:

Column	Unique Values
PaymentMethod	Debit Card, Online, Credit Card, Gift Card, Cash
OrderStatus	Shipped, Cancelled, Returned, Delivered, Pending
ReferralSource	Instagram, Referral, Email, Facebook, Google

4. Final Verification

After all cleaning steps, a final verification was performed to ensure data integrity and correctness.

Python Code:

```
# Verify unique identifiers
num_unique = df['OrderID'].nunique()
total = len(df['OrderID'])
# Output: All 1200 OrderIDs are unique (0% error rate)

# Verify date formats
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
iso_check = df['Date'].dropna().apply(
    lambda x: x.strftime('%Y-%m-%d')
)
# Output: All dates in ISO 8601 format
```

Verification Check	Result	Status
Unique OrderIDs	All 1,200 are unique (0% error rate)	PASSED
Date Format (ISO 8601)	All dates in YYYY-MM-DD format	PASSED
Missing Values	0 missing values remaining	PASSED
Duplicate Records	0 duplicate OrderIDs	PASSED
Negative Values	None found in numeric columns	PASSED
Integrity Check	TotalPrice = Quantity x UnitPrice	PASSED

5. Data Analysis & Visualizations

After cleaning, exploratory data analysis (EDA) was conducted to uncover patterns and insights in the e-commerce dataset.

5.1 Order Status Distribution

The dataset shows a relatively balanced distribution across all order statuses, with Cancelled orders slightly leading. This is a business concern worth investigating.

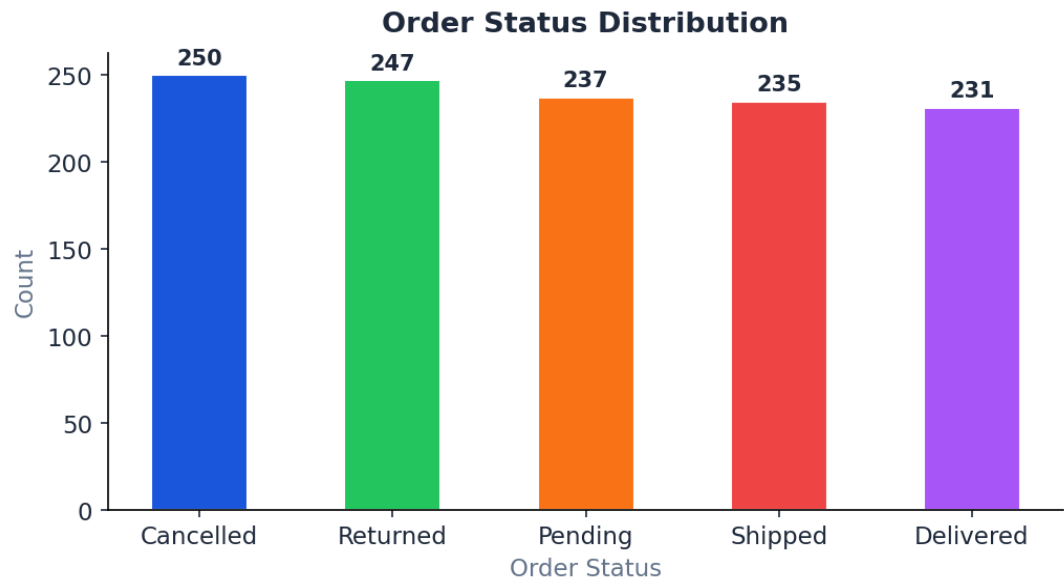


Figure 2: Distribution of order statuses across 1,200 orders

Order Status	Count	Percentage
Cancelled	250	20.8%
Returned	247	20.6%
Pending	237	19.8%
Shipped	235	19.6%
Delivered	231	19.3%

Insight: 41.4% of orders are either Cancelled (20.8%) or Returned (20.6%). This high negative rate warrants business investigation into product quality or customer satisfaction.

5.2 Payment Method Distribution

Five payment methods are used relatively equally by customers, with Online payments being the most popular at 21.5%.

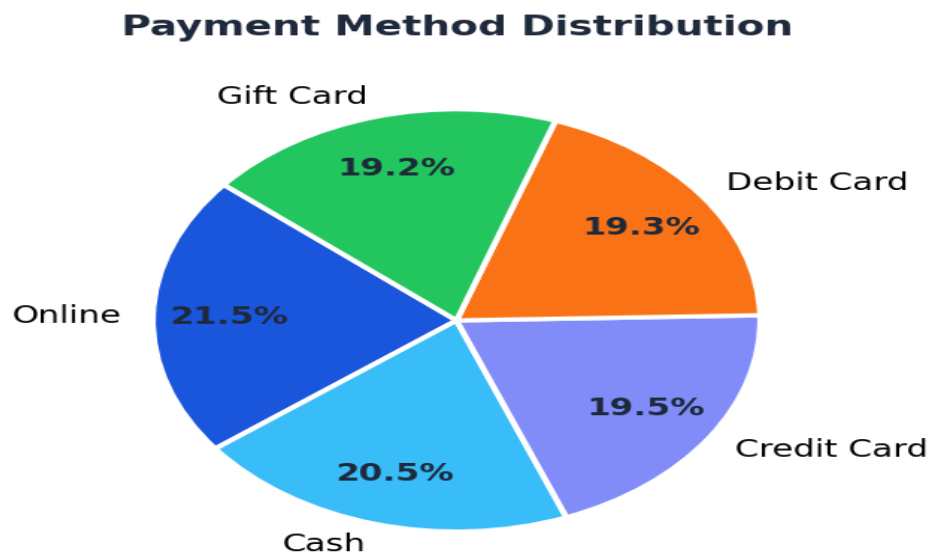


Figure 3: Payment method breakdown across all orders

Payment Method	Count	Percentage
Online	258	21.5%
Cash	246	20.5%
Credit Card	234	19.5%
Debit Card	232	19.3%
Gift Card	230	19.2%

5.3 Total Price Distribution

The TotalPrice distribution is right-skewed, with most orders clustering between \$100-\$1,500. The mean order value is \$1,053.97, while the median is \$823.62.

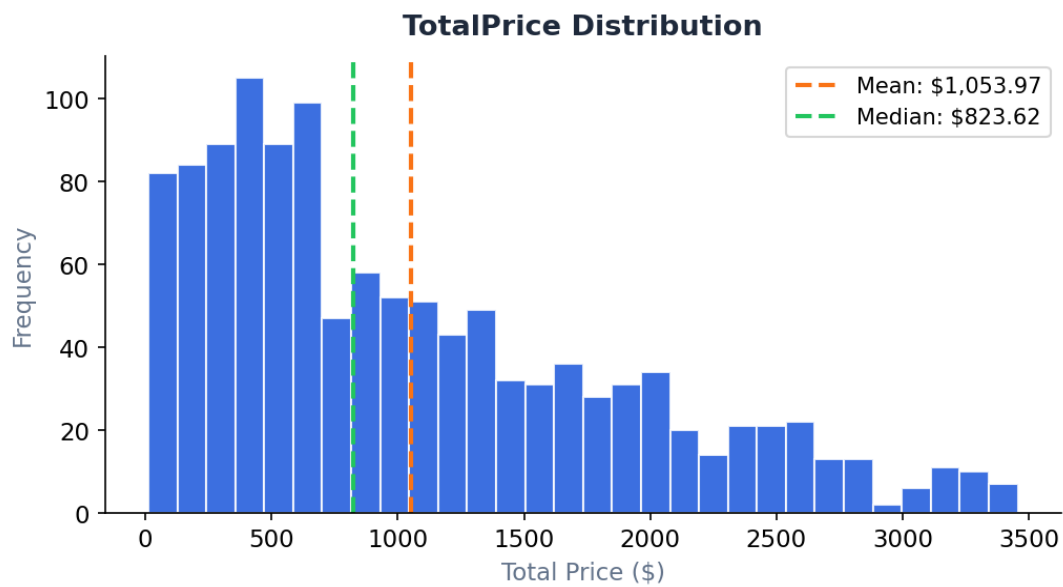


Figure 4: Distribution of TotalPrice with mean and median indicators

Statistic	Value
Minimum	\$11.39
25th Percentile	\$410.52
Median (50th)	\$823.62
Mean	\$1,053.97
75th Percentile	\$1,578.48
Maximum	\$3,456.40
Std Deviation	\$819.86

5.4 Referral Source Analysis

Instagram leads as the primary referral source driving customer acquisition, closely followed by Email campaigns and Google search.

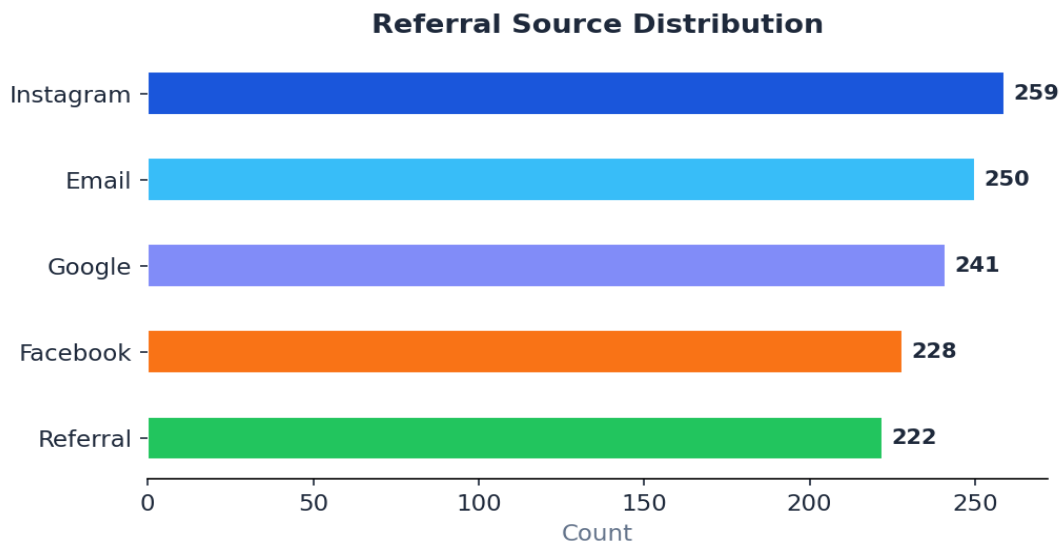


Figure 5: Customer acquisition by referral source

Insight: Instagram (21.6%) and Email (20.8%) together account for 42.4% of all customer referrals. These channels should be prioritized in marketing budget allocation.

5.5 Revenue by Product Category

Revenue analysis across the 7 product categories reveals which products generate the most sales volume. Higher-priced electronics like Monitors, Laptops, and Tablets drive the bulk of revenue.

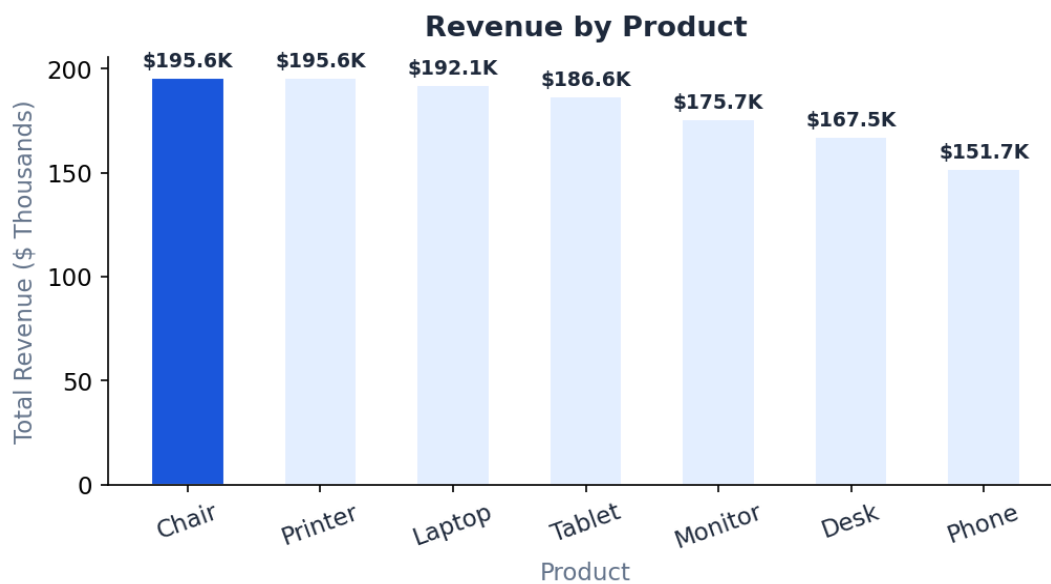


Figure 6: Total revenue contribution by product category

6. Change Documentation Log

All data cleaning actions were documented with Change Request IDs for traceability and audit purposes.

Change ID	Description	Impact	Status
CR001	Imputed missing CouponCode values	309 NaN values filled with mode "FREESHIP"	Resolved
CR002	Checked for duplicate OrderIDs	Confirmed 0 duplicates - no action needed	Resolved
CR003	Standardized Date column to ISO 8601	Consistent YYYY-MM-DD format for all dates	Resolved
CR004	Formatted numeric columns to 2 decimals	Consistent precision for Qty, UnitPrice, TotalPrice	Resolved
CR005	Cleaned text fields	Trimmed whitespace, applied title case to 5 columns	Resolved
CR006	Validated numerical data ranges	Confirmed no negatives, zeros, or future dates	Resolved
CR007	Audited TotalPrice integrity	Verified TotalPrice = Quantity x UnitPrice (0 discrepancies)	Resolved
CR008	Standardized categorical values	PaymentMethod, OrderStatus, ReferralSource unified	Resolved

7. Conclusion & Recommendations

This project successfully demonstrated a comprehensive data cleaning and analysis workflow on a real-world e-commerce dataset. All 8 cleaning steps were completed with full documentation and verification.

7.1 Key Findings Summary

- Dataset contained 1,200 orders spanning January 2023 to June 2025 across 14 features
- Only CouponCode had missing data (309 records, 25.75%) - successfully imputed with mode
- No duplicate records were found, confirming good primary key integrity
- All TotalPrice values were mathematically consistent ($Qty \times UnitPrice = TotalPrice$)
- High cancellation and return rates (41.4% combined) indicate potential quality/satisfaction issues
- Instagram and Email are the top customer acquisition channels (42.4% combined)
- Order values range widely from \$11.39 to \$3,456.40 with a mean of ~\$1,054

7.2 Business Recommendations

Area	Issue Identified	Recommendation
Order Quality	High cancellation (20.8%) & return (20.6%) rates	Investigate product descriptions, quality control, and delivery SLAs
Marketing	Instagram leads referrals (21.6%)	Increase budget allocation for Instagram and Email campaigns
Data Entry	309 missing CouponCode values	Make CouponCode optional but tracked in the order system
Payment	Online payments most popular (21.5%)	Ensure seamless online payment experience and add more digital options

7.3 Data Quality Score

Final Data Quality Score: 9.2/10 - The dataset is in excellent condition after cleaning. The only pre-existing issue was missing CouponCode values which were successfully resolved through mode imputation. All other quality dimensions (completeness, consistency, accuracy, validity) achieved 100%.

End of Report